

The main pattern: remove repeated work and duplicated authority

P9 · binder freshness

Scan preceding definitions
for each chronological law fill



Compute one binder bound;
keep visibility environments separate

4 / 8 / 16 fills: full-book scans 5 / 9 / 17 → 1 / 1 / 1

P16 · compact literals

Expand literals into constructor trees;
copy, freshen and encode the trees



Keep scalar/string payloads compact;
expose a constructor view on demand

Same-source census: 2,171,045 → 152,620 freshened terms (−92.97%)

P22 · contextual frontend

Parse first; reconstruct scope later;
repeat completion and diagnostics



Resolve at the right parsing checkpoint;
one contextual completion authority

Main: 2 → 0 differences; broader: 57 → 0; net compiler source: −300 lines